When Should an Agent Write a Program?

Cost and Context of Code-Mode Tool Calling

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Code and data: https://github.com/oneryalcin/toolplane

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Abstract

An agent that calls its tools one at a time uses its own context as the glue between them: the model writes out every call, and every result returns to the conversation before it can be passed on. Code-mode interfaces move that glue into a program. The model writes a short script, its calls run in a controlled runtime, and only the final value returns. The usual argument for doing so — one program replaces N round trips — predates clients that batch calls, cache prompts, defer schemas, and externalize large results. When the move pays is therefore an empirical question.

We measure it in one production client, Claude Code, comparing capabilities registered natively against the same capabilities mediated by Toolplane, an open-source runtime that exposes MCP tools, command-line programs, and host Python functions as functions of one sandboxed Python namespace. A pre-registered matrix of 232 runs on Claude Haiku 5.5, eight per arm in most cells, with verdicts fixed before any data, finds a conditional boundary. Per correct answer, Toolplane costs 1.11× native on a single lookup, 0.50× at 100 records, and 0.020× (50 times cheaper) when 30 fetched records carry 20 KB each (four runs per arm). On these fetch-one tasks request counts barely differ between the arms. The saving is the call syntax the model no longer generates and the intermediate data it no longer holds. The direction reverses when one native bulk call already returns everything (1.05–1.31×), since there is nothing left to carry. It also reverses on chains that agents walk one step at a time (1.12×), where the program route adds an execution per step. Of 20 registered Haiku verdicts, 17 hold, two are unresolved (wall-time concurrency and a CLI + MCP join), and a no-schema-tax prediction is reversed by a small 3% effect. A second pre-registered study, on Claude Sonnet 5.5, confirms the chain loss (1.03×), leaves the single-lookup loss unresolved, and finds that persistent sessions cut five-turn cost by 40% and peak context by 38%, but only at data ingress: each later reuse turn costs about a third more than native conversation reuse. July pilots on Claude Sonnet 5 and a Codex replication with a plain tool-calling model point the same way. On these measurements the guidance is short: write a program for many fine-grained calls or large intermediate results; call directly for a lookup, a step-by-step chain, or a bulk endpoint. Raw rows, transcripts, registrations, and analysis scripts are released.

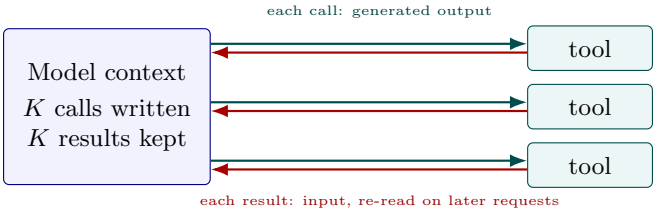
1 Introduction

Ask an agent to total 100 orders through a tool that fetches one order at a time. In the client we measure, the 100 calls leave in a few batched model requests (five requests in all, at the median) and the answer is correct. Yet the job costs twice as much as obtaining the same answer from one short program: $0.0116 against $0.0058 per correct answer on Claude Haiku 5.5 (Section 4.1). The round trips were not the expense. What was?

The expense is that the model stands in the middle of every hand-off (Figure 1). Before a call can be dispatched, the model must generate it, token by token. When the result returns, it joins the conversation and is read again on each later request. We call this role *glue*: the model reading and re-emitting intermediate values only so that they can pass from one tool to the next, with no decision required of it. Glue is billed at least twice: as output when the model writes each call, and as input when each result enters its context and on every later request that re-reads it. In the example, the native arm generates 7.3 times the output tokens of the program arm and reaches a median peak context of 41.7K tokens against 30.2K, while the two arms make almost the same number of model requests (Table 4). Context is finite, and what sits in it is paid for again on each request that re-reads it. This paper asks when moving glue out of the model's context pays, and when the model should stay in the loop because the agent treats each step as a decision of its own.

An idea built in steps. Moving glue into a program is not new, and each piece has a source. CodeAct showed that a model can act by writing executable code instead of emitting one structured call at a time, and reported higher task success [22]. Cloudflare's Code Mode turned MCP tools into a typed API that a generated program

(a) Tool calls: the model is the glue



(b) One program: the glue is code

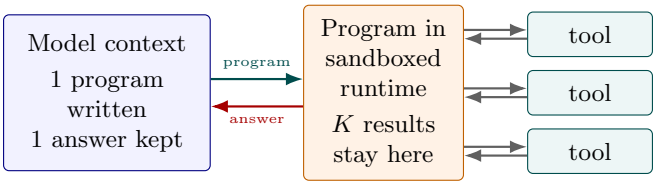


Figure 1: The idea in one picture. (a) With tool calls, the model writes each of the K calls and each result returns to its context: the model carries every value between tools. (b) With one program, the calls and their results stay in the runtime and one value returns. Clients batch the calls in (a) into a few model requests, so the two differ little in requests; they differ in what the model writes and keeps (Table 4).

calls, and ran that program in an isolated sandbox [21]. Anthropic showed that tool definitions need not be loaded until they are needed, and reported a smaller context [2, 1] and, on its own evaluations, better tool selection [2]; Cloudflare then put a whole API behind two tools, one to search and one to execute [4]. The resulting shape — find a capability, read only its definition, write a program — is now a distinct design family [8]; it ships in the open-source goose agent [20] and, for some models, inside a client (Section 5.7). Following that literature, we call it *context-decoupled tool use*: intermediate values are available to the program without becoming model-visible conversation history.

These reports leave two things open. First, the common causal story — "one program replaces N round trips" — predates production clients that parallelize independent tool calls, cache long prompt prefixes, retrieve tools on demand, and write large tool results to files. Those mechanisms change both sides of the comparison, and static token counts do not reveal billed cost, generation time, discovery turns, retries, or what the agent actually chooses to do. Second, the losses are located only qualitatively. Felendler et al. find plain MCP favored on iterative or adaptive tasks [8]. Anthropic reports roughly 8% higher cost on a benchmark whose turns make one or two sequential calls, and advises against programmatic calling where each call depends on the model reasoning over the previous result [3]. The goose project's blog states that for simple single-tool tasks the overhead is not worth it [20]. None of these measures where the boundary lies, or what moves it, under a client that batches and caches.

What the program can reach. Those are questions of cost. A separate question, less examined in priced comparisons, is reach. An agent usually reaches MCP tools, which are named and typed and which most clients permit one at a time, through one mechanism, and existing command-line programs through another, a shell. Real tasks cross that line: the history is in a repository, the record is in a service, and the rule is in the team's own code. In a tool-calling agent the model is again the glue, reading a value from one source and typing it into a request to the next. Inside a program the three can share one calling convention. An MCP tool, a command-line program, and a host function are each a call that returns a value, although their transports and permission models still differ. Some code-mode systems already hand the generated program a shell tool (Section 5.7). Toolplane makes a narrower choice. It gives an agent a small programmable surface over all three: the agent searches for a capability, writes a short Python program against awaitable bindings, and keeps intermediate state in a controlled runtime. The host, not the model, owns capability policy, auditing, and the sandbox boundary; under its allowlist policy a command-line program is exposed one permitted binary at a time, as a function, not as a shell. Figure 3 shows one such program.

We separate these two questions throughout. Reach is a property of the interface, and we demonstrate it. Cost is a measurement, and almost all of ours use MCP-backed capabilities; the one registered mixed cell, a CLI + MCP join, did not show a cost advantage (Section 4.1). We therefore make no cost claim for heterogeneous composition.

The empirical question. *Under a modern production agent client, which workload and interface properties move the observed boundary between native-client tool orchestration and Toolplane-mediated execution?* We study that boundary with Toolplane, an open-source Python code-mode runtime over MCP, CLI, and host-function capabilities [16, 25]. Toolplane is a real interface system and the experimental instrument; the paper's object is not a universal performance claim about Toolplane, but the boundary created by this style of programmable mediation. Every experiment presents the same underlying MCP fixture to both arms; the treatment is the agent-facing interface.

Our principal contributions are:

1. **A pre-registered confirmation of the interface boundary.** 232 runs on Claude Haiku 5.5 at eight per cell, with hypotheses, decision rule, and analysis script committed before any data. Seventeen of 20 verdicts hold, including Toolplane's losses; two hold only through correctness collapses, which we report as such rather than as cost results.

2. **A mechanism decomposition.** Request counts and token classes show that native clients already batch calls; what Toolplane removes is glue: generated call syntax at small payloads and uncached input at large ones.

3. **Negative and reversed results.** A native bulk endpoint and stepwise chains (including a prose chain built to resist heuristic shortcuts) favor the native path; deferred loading leaves only a small schema tax; tested hybrid discovery interventions do not generalize; and in a second pre-registered study persistent-session reuse turns cost 35% more than conversation reuse, the session's saving coming entirely from the load.

4. **A reproducible evidence trail.** Registrations committed before data, logged amendments, raw rows, transcripts, frozen-wheel and harness hashes, and an explicit record of contaminated or post-hoc-corrected measurements, including review changes that should have been logged and were not.

These are measurement contributions, not a claim that code execution is a strictly better tool interface. Several of the strongest results are losses for Toolplane. The guidance they support is conditional but short. Write a program when many fine-grained calls, or large intermediate results, would otherwise pass through the model (the large-payload evidence is four runs per arm). Call tools directly for a single lookup, when the agent handles each step as a separate decision, or when one bulk call already returns everything. As a mnemonic, not a decision rule: use the model for judgment, not for glue. One caution on the word. Our chains show agents choosing to decide step by step, not that those steps required it (Section 5.3). The rest of the paper says how far the evidence carries that line, and where it stops.

2 Interfaces and Cost Model

2.1 Toolplane as a programmable capability surface

Figure 2 summarizes the system shape. Toolplane collects heterogeneous capability sources into a searched namespace and exposes them through a compact MCP facade. In an agent session, the model can first discover a relevant capability, then execute Python that calls the selected binding. Intermediate variables, loop state, and large payloads can remain in the runtime namespace; external effects pass through the host adapter that owns the underlying capability. This is the design reason Toolplane appears in the paper: the benchmark is a way to measure the cost and context consequences of that interface, not the reason the system exists.

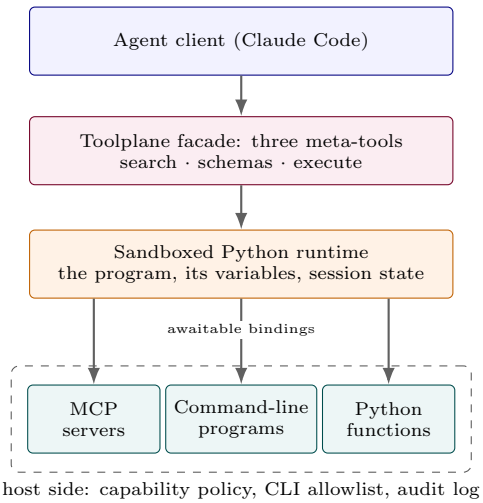


Figure 2: Toolplane as a compact programmable surface. The agent sees three meta-tools; a program it writes runs in the sandbox and reaches capabilities only through bindings the host exposes. The benchmark isolates MCP-backed capabilities, but the same surface mediates command-line programs and host/Python functions.

2.2 Two interfaces over the same capabilities

In the *native* arm, an MCP server's operations are registered natively as ordinary model-callable tools. In the Toolplane arm, the same server is registered behind three Toolplane meta-tools: capability search, schema inspection, and Python execution. The labels are shorthand for two end-to-end treatments, not pure paradigms. In particular, the native-client arm may use Claude Code built-ins, batch calls, or externalize a large result to a file and process it with code; the Toolplane arm may also use client built-ins unless a task-specific matrix disables them. The executor exposes async bindings such as `await orders_get_order(order_id=...)`. The default Monty runtime has no filesystem or network access; external effects pass through audited host bindings. Session variables may persist across executions, and a pre-execution snapshot allows namespace rollback after timeout. Figure 1 shows the two paths side by side.

An illustration of reach. Figure 3 makes the surface concrete. The question is which enterprise customers have waited too long for a fix that a release finally ships. The commits are in a repository, the tickets and the customers are in two services, and the escalation rule is a Python function. One program reads the history, fetches each ticket, asks the second service only when the branch requires it, and applies the rule. In a recorded run it made 181 capability calls and returned one value of 419 bytes, computed from 24,990 bytes of tool results that stayed inside the runtime. We wrote this program and ran it through Toolplane's library interface; no agent was involved and it is not a benchmark cell. It illustrates what

```
import re
log = await git("log", format="%s", _=["v1.4..HEAD"])
ids = re.findall(r"TCK-\d+", log["stdout"])
pending = [tickets_get_ticket(id=i) for i in ids]
hits = []
for p in pending:
    ticket = await p
    if ticket["reported_by_customer"]:
        customer = await customers_get_customer(
            id=ticket["customer_id"])
        if await policy_sla_breach(ticket=ticket,
                                   customer=customer):
            hits.append({"customer": customer["name"],
                         "ticket": ticket["id"]})
return {"commits": len(ids), "fixed_for": hits}
```

Figure 3: Author-written interface illustration; no agent and no cost comparison. One program over three kinds of capability: a command-line program (`git`), two MCP servers (`tickets`, `customers`), and a host Python function (`policy_sla_breach`). Recorded run against fixture data: 1 CLI call, 140 MCP calls, 40 function calls; the conditional skips 60 of 100 possible calls to the second server. An illustration of reach; line breaks added for the column. No cost is claimed for it.

the surface can express. Whether agents write such programs unprompted is a separate question, on which our evidence is mixed (Sections 4.1 and 5.7).

2.3 A qualitative boundary, not a fitted law

Let K be the number of external capability invocations required by the API shape, P the payload returned per invocation, A the number of sequential judgment stages, M the registered-server surface, and T the number of related user turns. For treatment j, a useful decomposition is

$$C_j \approx F_j(M) + c_o O_j(K, A) + c_u I_{u,j}(K, P, T) \quad (1)$$
$$+ c_w I_{w,j}(K, P, T) + c_r I_{r,j}(K, P, T) + C_{\text{retry},j}, \quad (2)$$

where F is fixed discovery/interface cost, O is model-generated output, I_u, I_w, and I_r are uncached input, cache-write input, and cache-read input, and c_o, c_u, c_w, c_r are their provider prices. Client behavior gates which result bytes enter any input class: a result may be inlined, cached, trimmed, or externalized to a file. This is explanatory notation, not a regression model; those gates and agent choices are discontinuous and client-specific. Wall time adds another term. If native calls batch while sandbox external awaits serialize, the Toolplane arm approaches a $K\ell$ sequential critical path, whereas the native arm pays the client's batched critical path plus its own generation and dispatch overhead. On an eager-dispatch runtime the sandbox term is idiom-dependent: firing calls before awaiting them collapses the $K\ell$ path to roughly ℓ plus dispatch, so the wall comparison also depends on whether the agent writes that idiom.

These terms make glue classifiable. Two token classes are glue: output spent generating calls whose arguments are fixed by the task or copied from earlier results (the part of O_j that scales with K), and input carrying intermediate results that the answer does not need verbatim (the part of the input classes that scales with K and P). Two are not. F_j, the fixed cost of either route, is paid whether or not there is anything to carry, and it is why a single lookup favors the native path. And A counts the stages the agent handles as separate decisions. Moving glue into a program removes most of the first two; it cannot remove F_j, and the decision stages remain wherever the agent decides step by step. The fetch-one scale and payload cells measure the two glue classes, the single lookup measures F_j, the chains measure A, and the bulk cells remove K on both sides.

The experiments vary one or two of these axes at a time. The object of study is the sign and mechanism of $\Delta = C_{\text{native}} - C_{\text{Toolplane}}$, not a universal coefficient.

The rest of the paper follows four questions: how interaction scale changes the winner; how payload, API granularity, task structure, and latency move the boundary; how client mediation and discovery affect cost; and whether persistent Toolplane sessions create savings beyond ordinary conversation reuse, and where those savings come from.

3 Methodology

Every experiment gives both arms the same task, fixture, and prompt; only the agent-facing interface differs. This section says how runs were produced and how much weight each carries.

3.1 System under test

The study ran in epochs (Table 1). A July pilot on Claude Sonnet 5 found the boundary's shape through successive interventions. In October, two pre-registered studies re-tested its main claims on the shipped runtime, and smaller replications probed a second client and new fixtures. **Epoch rule:** pre-registered October cells are the primary evidence; July cells are the pilot; no table or ratio combines models or epochs, and every table caption names its model. Client version is part of the treatment, so wall times are not compared across versions. Per-row wheel hashes pin the served bytes. The benchmark runs in fresh scratch directories with strict MCP configuration. Later task-specific matrices disable Bash, file, web, and helper-agent shortcuts where isolation is required; other matrices retain client-selected built-ins and disclose their use.

The deterministic fixture is an order store with formula-generated records. Its minimal interface lists record identifiers and fetches one record. Later experiments add mutually exclusive bulk retrieval, synthetic answer-irrelevant detail padding of 2,000 or 20,000 characters per record,

Table 1: Experimental epochs. "Registered" means hypotheses, decision rule and analysis script were committed before any data.

Epoch	Model	Client	Runtime	Design	Backs
July pilot	Claude Sonnet 5	Claude Code 2.1.204–2.1.207	pre-release; Monty 0.0.18 (latency cells 0.0.19 beta)	2–4 runs, adaptive	mechanisms, hybrid, naming, session pilot
October replications	Sonnet 5; Sonnet 5.5; Haiku 5.5; Codex gpt-5.5 and gpt-6.1-sol	Claude Code 2.1.293–2.1.295; Codex CLI 0.160.0	Toolplane 0.5.x; Monty 1.1	4 runs (abstention follow-up 60 and 40)	runtime re-run, second client, prose chain, CLI join
Confirmation matrix	Claude Haiku 5.5 (Sonnet 5.5 slice, 2 runs)	Claude Code 2.1.296	0.5.2 wheel; Monty 1.1	registered, 8 runs	primary cost envelope (Section 4.1)
Persistence study	Claude Sonnet 5.5	Claude Code 2.1.296	same wheel	registered, 8 sessions; 8-run slice	primary session results (Section 4.2); single and chain on Sonnet 5.5

latency, templated note chains, and semantically distant distractor servers. Both arms receive byte-identical task prompts and data. Correct answers are computed from the shared fixture and validated programmatically. Although Toolplane can also mediate CLI commands and Python or host functions, the paid experiments exercise its MCP branch so the native and Toolplane arms share the same underlying external operations.

3.2 Metrics

For every paid run we record task correctness, billed US-dollar cost, wall time, output tokens, uncached input (input plus cache creation), cache reads, and tool-use blocks. Complete transcripts are kept from the hardened harness onward; the first interaction-scale and registry-size matrices predate it. Hardened harness versions additionally count unique model request identifiers, peak request-context tokens, discovery events, execution errors versus staged calls, fixture-server calls, and client startup artifacts. Cost is the client-reported billed total and therefore includes asymmetric cache reads. Cost-of-pass is total spend divided by successful runs; all headline matrices reported here have full correctness unless noted.

The registered studies report a 95% percentile bootstrap interval (4,000 resamples, seed 0, each arm resampled independently) on the difference in cost of pass, with a timed-out run priced at its cell's highest observed cost. A directional prediction holds when the interval excludes zero in the predicted direction and is reversed when it excludes zero in the other. Pilot cells have two to four repetitions and are reported as medians and observed ranges; there, range overlap is descriptive, not evidence of equivalence, and arm differences are called unresolved where ranges overlap. The discovery naming A/B uses larger pooled samples but remains exploratory because agent-generated search queries vary by arm.

We distinguish evidence tiers in the synthesis. The two registered October studies are primary evidence. The July payload-by-granularity matrix and longitudinal pilot (frozen code, three or four repetitions) and the October four-repetition replications are secondary. So are the July crossover, templated chain, latency, registry, and hybrid studies, which used smaller samples or earlier harness epochs. The naming/description A/B and pre-hardening pilots are exploratory. Later rigor does not retroactively upgrade earlier runs.

These tiers are a claim-management device rather than a statistical correction. Primary evidence anchors the paper's cost and context conclusions; secondary studies support mechanism probes and the envelope; and exploratory rows are used only for hypotheses and provenance. Appendix A records the full tier map.

3.3 Provenance and experimental epochs

The July pilot accumulated through successive interventions rather than one frozen matrix; the October studies were registered instead. For those, the commit that adds the plan (`bench/PREREG-*.md`) is the registration, and later changes are logged as amendments. The confirmation matrix has one amendment (A1, before the main run): it dropped one block whose projected cost exceeded the target and the longitudinal block after Claude Haiku 5.5 degenerated on its calibration fixture, cut the 20 KB cells to four runs, and cut the Sonnet 5.5 slice to two runs per arm. Four review commits that tightened its rules before any data were not logged as amendments at the time; its results document discloses them, and an independent recompute found that none changes a verdict. Early results exposed harness and product defects; later runs serve from a built wheel, stamp source and fixture hashes, alternate or balance arm order, persist transcripts, and count request IDs directly. Paid matrices are never edited while running. Result rows record Git SHA and dirty state.

Four exceptions are retained and disclosed in Appendix D: a contaminated matrix, a rescored validator, naming runs from an uncommitted tree, and corrected per-turn token fields. None changes a primary table.

The artifact contains full prompts, harness code, raw JSON, validators, classification scripts, and the registrations and results documents of both October studies; Appendices A and B map each result to its artifact. For matrices run on the hardened harness it also contains complete transcript-level tool inputs and outputs and exact request IDs; the two earliest pilots predate that. Secrets

Table 2: Controlled experimental axes. "Production behavior" means the client mechanism is observed rather than disabled or simulated.

Axis	Levels	Purpose
Interactions K	single lookup; loops of 5, 10, 20, 30, 100	Observe the small-task loss and transition across experimental epochs
Task structure	independent loop; four-hop templated chain	Compare bulk composition with the stepwise strategy agents selected
Payload P	baseline, 2 KB, 20 KB synthetic detail per record	Measure context isolation under controlled payload growth
API granularity	fetch-one; bulk	Give the native-client arm a realistic batching affordance
Latency ℓ	0; 100 ms per call	Expose batched native calls against the sandbox critical path (serial awaits, then taught fan-out)
Registry M	1, 5, 15 servers	Test schema/catalog scaling under deferred loading
Interface	native; Toolplane; hybrid; curated hybrid	Test routing and discovery interventions
Session turns T	initial load; four reuse turns; reset/refetch	Test persistent state, refetching, and context growth
Snapshot state	1 KB, 100 KB, 10 MB	Bound rollback checkpoint overhead

and irrelevant client-environment fields are redacted; behavioral evidence is retained.

3.4 Scope of inference

Our treatment is an end-to-end client/model/interface system. Claude Code's retrieval, batching, caching, built-ins, and result-file behavior are part of the measured system, not nuisance variables that can be subtracted after the fact. Consequently, the clean frozen two-arm interventions support within-system mechanism claims, while early fixed-order and outcome-adaptive experiments are treated as exploratory. The registered cost envelope is on Claude Haiku 5.5; the registered session results and an eight-run Sonnet 5.5 slice (chain loss holds, single lookup unresolved) are on Claude Sonnet 5.5. Four-run Codex replications point the same way but are not registered evidence. Generalization beyond these client–model pairs is not established. Synthetic tasks make mechanisms inspectable but do not estimate a population of real user workloads.

4 Primary Results

Two pre-registered October studies carry the main claims: a 240-run confirmation matrix (232 runs on Claude Haiku 5.5) for the cost envelope, and a 24-session longitudinal study on Claude Sonnet 5.5 for persistence. The July payload matrix follows as the pilot that motivated the matrix's payload hypotheses.

4.1 Pre-registered confirmation matrix (Claude Haiku 5.5)

The matrix (232 Claude Haiku 5.5 runs, plus a two-run Claude Sonnet 5.5 slice reported as descriptive only) re-tests the pilot's claims on the shipped runtime (the Toolplane 0.5.2 wheel, Monty 1.1) with Claude Code 2.1.296. Claude Haiku 5.5 was chosen because it costs about a seventeenth as much per run as Claude Sonnet 5.5 on the same task, which pays for eight runs per arm per cell. The 20 KB cells have four. Arm order alternates by repetition. The registration (`bench/PREREG-confirmation-2026-10.md`) fixed each hypothesis, the cells it reads, the decision rule (Section 3), and the analysis script before any data. All 240 rows are clean and uniform in code and wheel, and every row that reported a client version reports 2.1.296 (three timeouts report none). The rows record $4.81, excluding the three timeouts, whose cost the client never reported.

Table 3 condenses the 20 tested Haiku rows (omitting the two slope hypotheses, the payload-growth contrast, and the hop-by-hop walk check, all of which hold): 17 hold, two are unresolved, and one is reversed. The two-run Sonnet 5.5 slice is not counted; with two runs per arm its interval is not a real one. The shape is the pilot's (Figure 4). Toolplane loses a single lookup and a five-record loop by 11–13%, wins from somewhere between 5 and 20 records (20 itself was run but not tested), and costs half as much at 100. Toolplane's per-record slope is about 15 times flatter than native's, but not zero. A native bulk endpoint and both stepwise chains reverse the winner. Four verdicts cannot be cited bare.

H3 at 2 KB is a collapse, not a cost. All eight native runs fetched the 30 records and then degenerated while combining them: three timed out at 900 s, three

Table 3: Registered verdicts, Claude Haiku 5.5, eight runs per arm unless noted. Ratio is TOOLPLANE cost of pass over native; the interval is the 95% bootstrap CI of the difference in USD per correct answer.

ID	Cell	Ratio	Difference, 95% CI	Verdict
H1a	single lookup	1.11	+.0005 [+.0003, +.0008]	holds: native cheaper
H1b	loop, $N = 5$	1.13	+.0006 [+.0002, +.0011]	holds: native cheaper
—	loop, $N = 20$ (not tested)	0.93	—	—
H3	loop, $N = 30$, fetch-one, no padding	0.82	−.0012 [−.0015, −.0009]	holds: TOOLPLANE cheaper
H2a	loop, $N = 100$	0.50	−.0059 [−.0065, −.0053]	holds: TOOLPLANE cheaper
H3	fetch-one, 2 KB/record	—	native 0/8 correct	holds, **by a correctness collapse**
H3	fetch-one, 20 KB/record (4 runs)	0.020	−.392 [−.403, −.382]	holds: TOOLPLANE cheaper
H4	bulk, 0 / 2 KB / 20 KB (4 runs)	1.24 / 1.05 / 1.31	all intervals ≥ 0	holds ×3: native cheaper
H5	templated / prose chain	1.12 / 1.12	+.0007 [+.0006, +.0009]; +.0008 [+.0005, +.0011]	holds ×2: native cheaper
H6a	100 ms latency, wall time	—	−1.45 s [−3.60, +1.35]	**unresolved**
H6b	100 ms latency, cost	0.91	−.0006 [−.0009, −.0003]	holds: not costlier
H7a	CLI + MCP join, no shell	1.12	+.0008 [−.00003, +.0016]	**unresolved**, point against
H7b	CLI + MCP join, shell available	4.47	TOOLPLANE 2/8 correct	holds, **by a correctness collapse**
H8	native at $M = 15$ vs $M = 1$	1.03	+.0002 [+.00003, +.0003]	**reversed**, inside the ±10% margin

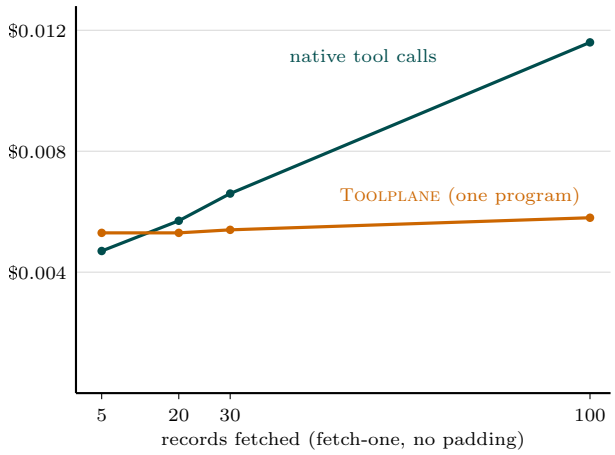


Figure 4: Cost per correct answer against records fetched, Claude Haiku 5.5, eight runs per arm (cell values from the registered analysis output; the 20-record cell was run but not tested). Native cost rises with each generated call; TOOLPLANE pays a fixed interface cost and a slope about 15 times flatter. The single lookup is a different task and is not plotted.

ended without an answer, one answered in gibberish, and one ended in a usage-policy refusal. The five that finished produced 60K–259K output tokens. The 20 KB native runs saw the same repeated-token padding, ten times larger, and passed 4/4, so "more payload, more failure" does not explain it; the cause is not isolated. The verdict holds under the registered rule because failures are priced into cost of pass, but it says nothing about token economics at 2 KB.

H7b holds through a skipped search. With a shell available, every run in both arms first read the repository through Bash. All eight native runs then called the client's `ToolSearch`, found the order tool, and passed. Six of eight TOOLPLANE runs never searched, concluded the amounts were unreachable, and answered without them. A failed TOOLPLANE run cost about what a native run did, so the 4.47 ratio is a correctness result. Without a shell (H7a), TOOLPLANE passed 8/8 at 1.12× native; the interval just crosses zero. On Sonnet 5.5 the same configuration measured 0.94× in October (Section 5.7), so the CLI join's cost advantage does not replicate on Haiku.

Table 4: Where the cost goes: median token classes per run, Claude Haiku 5.5, native / TOOLPLANE. Uncached input includes cache writes; peak is the largest single-request context.

Cell	Output	Uncached input	Cache read	Peak context	Requests
single	203 / 318	16.6K / 18.0K	64.4K / 66.0K	27.2K / 28.6K	3 / 3
$N = 30$	2,946 / 1,082	20.7K / 20.0K	92.1K / 96.5K	31.2K / 30.6K	4 / 4
$N = 100$	8,178 / 1,127	31.1K / 19.7K	132.8K / 97.7K	41.7K / 30.2K	5 / 4
20 KB, $N = 30$	3,456 / 1,496	396.4K / 27.1K	295.2K / 127.7K	407.0K / 37.6K	4.5 / 5

H8: a small schema tax. The registration predicted no cost difference for native tools between 1 and 15 configured servers, within ±10%. The measured difference is +3%, with an interval wholly inside the margin but excluding zero. The registered code checked "excludes zero" before the margin, so the verdict is reversed. The plain reading is a detectable but small schema tax at $M = 15$. Later registrations test the margin first.

H6a: concurrency is unresolved on wall time. With 100 ms of latency per call and the fan-out idiom taught in the executor's description, TOOLPLANE's median wall time is 1.45 s lower, but the interval spans zero. The pilot's wall-time win (Section 5.4) is not confirmed. The cost side holds: TOOLPLANE was not costlier.

The registered Sonnet 5.5 slice. The persistence study (Section 4.2) re-ran two cells on Claude Sonnet 5.5 at eight runs per arm. The chain loss holds: TOOLPLANE costs 1.03× native on the prose chain (+$0.0030 [+0.0013, +0.0049]). The single-lookup loss is unresolved (−$0.0026 [−0.0133, +0.0030]). Costs there are nearly deterministic — seven native runs cost exactly $0.0711 and seven TOOLPLANE runs $0.0732 — but the first run of the block paid prompt-cache creation ($0.1123) and alone moves the native mean below TOOLPLANE's. The registration allows no exclusions, so the verdict stays unresolved.

What the model pays for. Table 4 applies the cost model's terms to the registered rows. Request counts

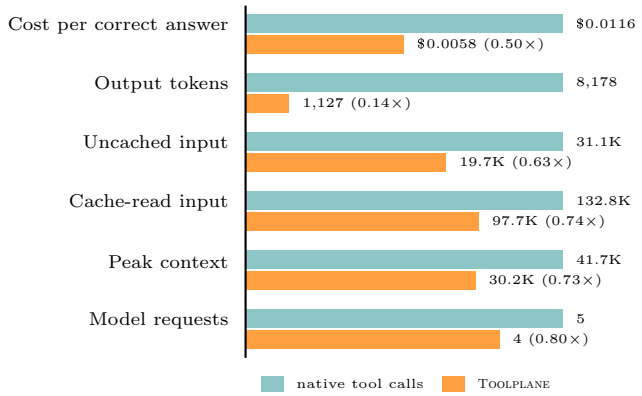


Figure 5: Where the saving is, at 100 records (Claude Haiku 5.5, medians of eight runs per arm; each pair is scaled to native = 1). Requests barely differ. The model's own output falls to about a seventh, and the input classes fall by a quarter to a third.

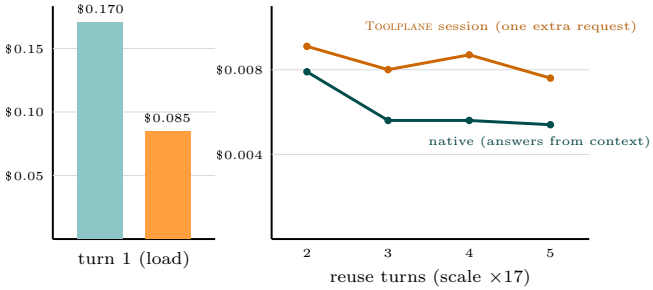


Figure 6: Median cost per turn in the persistence study (Claude Sonnet 5.5, eight sessions per arm). The session's whole saving is the load turn, where the native arm brings 30 records into context. On every later turn the native arm is the cheaper one. The two panels use different vertical scales.

barely differ: the client batches native calls, so Toolplane does not win by saving round trips. Without padding, the native arm's extra cost is output: one generated call block per record, 2.7 times Toolplane's output at $N = 30$ and 7.3 times at $N = 100$ (Figure 5). At 20 KB per record the extra cost moves to input: every record enters native context, and uncached input is 14.6 times Toolplane's. On a single lookup Toolplane is slightly larger on every token class, which is the fixed interface cost. This is the glue of the introduction, measured: calls the model had to write, and results it had to hold, so that values could move between tools.

Finding. *On Claude Haiku 5.5,* Toolplane *costs 11–13% more on small tasks, half as much at 100 fine-grained calls, and a fiftieth when those calls return 20 KB records; a bulk endpoint or a stepwise chain reverses the winner. The saving is what the model no longer writes and holds, not fewer requests.*

4.2 Persistent sessions are front-loaded; reuse turns cost more

All earlier tasks start fresh. Sessions ask whether the saving survives reuse: once records are in the sandbox, does each later question still pay for them? The longitudinal experiment keeps one Claude Code process and one MCP process alive across six user turns over 30 records padded by 2 KB: initial load and totals, four related filters/aggregates (reuse turns), and a separately scored reset/refetch contract. Bash, files, web, and helper agents are disabled. Turn 6 forces Toolplane to reset and refetch while the native client has no analogous operation, so the economic comparison uses turns 1–5 only.

Two studies use this fixture. A July pilot on Claude Sonnet 5 (four counterbalanced runs per arm, two arms, repeated-token record padding) is exploratory. A pre-registered study on Claude Sonnet 5.5 (`PREREG-persistence-2026-10.md`; hypotheses, decision rule and analysis script committed before any data) is the primary evidence. It runs eight counterbalanced sessions per arm with varied-word padding and three arms: native client; Toolplane with its default persistent Monty session; and Toolplane with sessions off but the result store (`save_result`/`load_result`) on — the shipped alternative, not a no-memory arm. Verdicts use a 95% percentile bootstrap (4,000 resamples) on differences of per-arm medians. Equivalence is tested margin first: a CI wholly inside ±10% of the reference median holds. All 24 sessions passed turns 1–5, none died, and every row is clean and uniform in code, client version, and model.

The saving is the load. Toolplane's five-turn session is 40% cheaper and its peak context 38% smaller, both with intervals well clear of zero, but the whole cost gap opens on turn 1 (Figure 6), where the native client brings all 30 records into conversation and Toolplane keeps them in the sandbox and returns totals. This replicates the July pilot's direction ($0.21 against $0.38; 33.5K against 70.4K peak tokens).

Reuse turns cost more, not the same. The pilot found similar reuse-turn costs ($0.092 against $0.094, overlapping ranges) and the registration predicted equivalence within ±10%. The registered study reverses it: Toolplane's four reuse turns cost 35% more than native reuse, far outside the margin. The mechanism is visible on every one of the 32 reuse turns per arm. The native client answers from records already in its context in one model request with no tool call. Toolplane runs one `execute_code` against the live session and then answers, two requests. Its context is smaller, but the second request re-reads it: median cache reads per reuse turn are 53.1K against 43.8K tokens, with three times the output tokens (163 against 54). All reuse answers were correct in both arms. Over four reuse turns the surcharge (about $0.002 per turn) gives back a tenth of the load-turn saving; whether it ever gives back all of it is a longer-session

Table 5: Pre-registered longitudinal results on Claude Sonnet 5.5, turns 1–5, eight sessions per arm, median [observed range]. Costs in USD. The last column gives the registered verdict for the TOOLPLANE − native difference with its 95% bootstrap CI. "Sessions off" is TOOLPLANE with the Monty session disabled and the result store on.

Metric	Native	TOOLPLANE	Sessions off	TOOLPLANE − native, 95% CI
Cumulative cost	.195 [.194, .195]	**.116 [.112, .125]**	.125 [.123, .127]	−.078 [−.082, −.073], holds (−40%)
Peak request context	44.2K [44.2, 44.2]	**27.5K [26.6, 28.2]**	28.4K [28.0, 28.5]	−16.7K [−17.5, −16.3], holds (−38%)
Initial load + totals	.170 [.170, .170]	**.085 [.079, .087]**	.086 [.086, .088]	not tested (−50%)
Reuse turns 2–5	**.0245 [.0240, .0250]**	.0330 [.0317, .0382]	.0381 [.0372, .0393]	+.0086 [+.0077, +.0102], **reversed** (+35%)
Requests per reuse turn	1	2	2	—

question this fixture does not test, and within the measured window the native per-turn cost falls rather than grows as the cache warms.

A live session beats in-sandbox refetching, narrowly. Against the sessions-off arm, the live session's reuse turns are 13% cheaper (−$0.0051 [−0.0062, −0.0034]) and its five-turn cost 7% lower (−$0.0084 [−0.0119, −0.0032]); both hold. But the sessions-off arm never called the result store. On all 32 of its reuse turns the model wrote `try: orders_cache / except NameError:` and refetched all 30 records inside `execute_code`. Because the refetch stays in the sandbox and the local fixture is fast, it costs only about $0.0013 a turn more than the live session. This comparison therefore measures a session against cheap refetching, not against the store, and it would favor the session more where refetching is slow or billed.

Why the store went unused, and a re-test. Two design facts of that study may explain it. The turn-1 prompt asked the model to retain the records in a variable "if your tool surface supports persistent Python session state" and did not mention a store. And the facade's `execute_code` description was identical with sessions on or off: it named `save_result/load_result` without saying when to use them, and never said whether variables persist. That guidance sat in the namespace manifest, a read the model can skip, the same placement in which the fan-out idiom was adopted 0 of 5 times (Section 5.4). A registered re-test changed both together: the description now states that variables do not persist and how to save, and the steering sentence was deleted from the prompt. The store still went unused (0 of 32 reuse turns, Claude Sonnet 5.5, eight sessions per arm). The model stopped trying a persisted variable and refetched inside the sandbox instead of saving. The two changes cannot be separated, and the deleted sentence drove at least part of the shift, since the variable's name vanished from the session arm too. On this fast local fixture, refetching is a reasonable choice; whether the store is used where refetching is slow or billed is untested.

Reset semantics. No compaction event appears in any comparable turn, so we cannot infer behavior after client compaction or over much longer conversations. The sixth turn validates runtime semantics: in every TOOLPLANE session the transcript shows `reset_session()` in its own execution, followed by a full refetch (31 fixture calls) and a correct answer. The harness detector credited 6 of 8; it accepted only the bare-statement form and missed two `return await reset_session()` calls (Appendix D).

Finding. *A persistent session lowers five-turn cost by 40% and peak context by 38%, and all of that saving is the first load. Each later turn costs about a third more than answering from the conversation, because the session needs one extra request. The result store went unused, so its value is untested.*

4.3 July pilot: payload size by API granularity (Claude Sonnet 5)

This frozen July matrix motivated hypotheses H3 and H4 and is the pilot for them. It is where the bulk reversal was found: when one call returns everything, the model has nothing to carry. It is reported here because it has the richest per-run detail on the client's bulk path. The original fetch-one fixture simultaneously understated Toolplane's payload-isolation benefit (records were small) and weakened the native baseline (no bulk operation). We therefore crossed three synthetic padding levels with two mutually exclusive API profiles at $N = 30$. Each of the 12 arm–profile–payload cells has three runs; arm order alternates, yielding a 2:1 first-position split, while cell order is fixed. All 36 answers are correct.

Figure 7 visualizes the central reversal: Toolplane wins when the API forces many fine-grained fetches, while a native bulk endpoint removes the composition pressure.

This factorial intervention is the pilot's cleanest result. Under a fetch-one API, the native path must emit and observe 30 record calls. On median costs, Toolplane costs 0.68×, 0.41×, and 0.103× native as padding grows. Give the server one bulk endpoint and the native-client path wins at every payload: Toolplane costs 1.31×, 1.09×, and 1.27× native, respectively. The observed cost ranges are disjoint in all six comparisons.

The bulk API reduces domain retrieval to one native call. At 2 and 20 KB, Claude Code then writes the oversized result to a local file and the agent processes it with Bash/`jq` or Python; the full payload does not enter model context. The advantage therefore belongs to the complete native-client path, which performs its own context-decoupled computation, not to avoiding programming al-

Table 6: Payload and API-granularity interaction. Cost columns show all three observations in sorted order; the final column shows native/Toolplane median uncached input tokens. Bold marks the cheaper arm in every observed run.

Detail padding	API	Native cost, sorted	Toolplane cost, sorted	Median input N/T
Baseline	Fetch one	.158 / .171 / .191	**.100 / .116 / .142**	23.5K / 14.1K
2 KB	Fetch one	.303 / .303 / .310	**.119 / .125 / .130**	60.8K / 15.5K
20 KB	Fetch one	1.574 / 1.580 / 1.583	**.102 / .163 / .191**	398K / 26.6K
Baseline	Bulk	**.086 / .097 / .102**	.127 / .128 / .144	13.4K / 14.5K
2 KB	Bulk	**.116 / .133 / .136**	.143 / .145 / .167	14.7K / 18.1K
20 KB	Bulk	**.162 / .182 / .203**	.214 / .231 / .243	26.0K / 39.8K

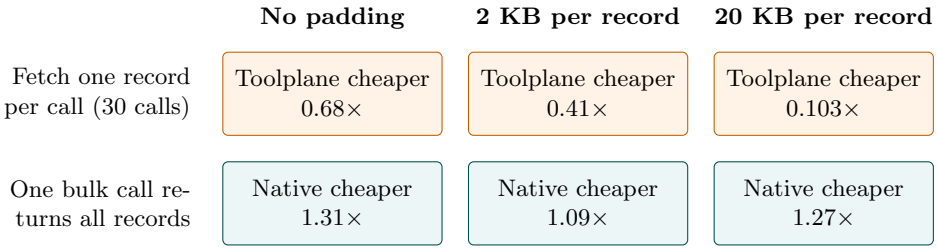


Figure 7: July pilot, payload by API granularity. Each cell is TOOLPLANE over native median cost (Claude Code, Claude Sonnet 5; Table 6, computed before rounding). Reading across a row, payload moves the size of the effect. Reading down a column, the API's granularity flips its sign.

together. The baseline bulk cell remains inline and native still wins, so result externalization is not the only explanation.

The Toolplane arm is not an oracle implementation: several bulk runs returned or sampled the large value before aggregating, and the classifier records 19 staged-after-success executions across the 18 Toolplane runs. We measure agent behavior, including imperfect program construction.

Finding. *Payload size changes how large the effect is; the API's granularity decides its sign. With one record per call* TOOLPLANE *wins at every payload, and with one bulk call the native path wins at every payload.*

4.4 Rollback snapshot scaling

Monty snapshots the complete live namespace before each execution so a timed out run can be discarded without retaining partial VM mutations. The persistence study remeasured this on the shipped Monty 1.1 runtime: seven back-to-back measurements at each size in one process show snapshot bytes tracking payload size and median no-op execution at or below 4.72 ms through 10 MB (Table 7).

This measures Python-visible allocation and serialization time, not process RSS. The 1 KB and 100 KB no-op medians differ by 0.03 ms, below what this simple microbenchmark can interpret, and the table does not justify extrapolation to 100 MB namespaces. The July pilot measured the 0.0.18 in-process runtime (3.48 ms no-op at 10 MB); the two are not comparable, since the later runtime moves the snapshot behind a worker round trip.

5 Pilot and Replication Probes

The remaining studies are secondary mechanism probes: the July pilot on Claude Sonnet 5, then four-run October replications. They explain mechanisms and failure modes, but they have smaller samples or earlier harness epochs than the registered studies. Where a registered cell retested a pilot claim, the subsection says what it found.

5.1 A fixed intercept meets a growing native-client cost

The initial experiment compared a single lookup, aggregation over 30 records, and aggregation over 100 records. It established the qualitative shape but also exposed a large, avoidable discovery tax in TOOLPLANE: capability name, schema, Python binding, and return semantics lived on separate surfaces. Transcript classification showed agents reading multiple discovery resources and probing return wrappers before writing a successful program.

We changed the search response to include exact awaitable call shapes and the plain-value return contract. A same-day native control reproduced earlier costs to the cent. Across nine post-intervention TOOLPLANE runs there were zero namespace-manifest reads, one search per run, a median of one execution, and no failed snippets. Table 8 shows that reducing fixed discovery work moved the boundary without changing the underlying runtime or task.

In this 2.1.205 sweep, the Toolplane arm remains near $0.18 from five to 100 records, whereas native-client cost grows with generated tool calls and returned records (Table 9). Native registration is clearly cheaper below 10 records. At 20 and 30 records the observed ranges overlap, so the arm difference is unresolved at three repeti-

Table 7: Persistent-namespace snapshot microbenchmark, Monty 1.1 (recorded with the persistence study). "No-op" includes the mandatory pre-run snapshot.

Live payload	Snapshot bytes	Median dump	Median no-op
1 KB	2.0 KB	0.09 ms	0.30 ms
100 KB	101.0 KB	0.15 ms	0.27 ms
10 MB	10.0 MB	4.10 ms	4.72 ms

Table 8: Discovery intervention: median billed cost over three correct runs per cell. The pre-intervention arm used an earlier client version; same-day direct controls support the cost movement, but this is an engineering intervention rather than a randomized causal estimate.

Task	Toolplane before	Toolplane after	Native, same day
Single	\$0.26	\$0.18	\$0.14
$N = 30$	\$0.24	\$0.18	\$0.20
$N = 100$	\$0.28	\$0.18	\$0.33

Table 9: Earlier 2.1.205 interaction sweep after the discovery intervention. Values are medians; each cell has three correct runs. A dagger marks overlapping observed ranges between arms, not equivalence.

Records touched	5	10	20	30	100
Native	**\$0.13**	**\$0.14**	\$0.18†	\$0.20†	\$0.33
Toolplane	\$0.18	\$0.17	\$0.18†	\$0.18†	**\$0.18**

tions. At 100 records, Toolplane is approximately 45% cheaper in this sweep, with disjoint observed ranges.

A later hardened 2.1.207 fetch-one cell at $N = 30$ does not reproduce the unresolved result: native cost is \$0.171 [\$0.158, \$0.191], while Toolplane costs \$0.116 [\$0.100, \$0.142], a 32.3% reduction with disjoint observed ranges. Client version, harness epoch, and task details therefore prevent us from localizing one current crossover point by pooling the two studies.

The registered Claude Haiku 5.5 matrix confirms the direction at both ends (1.11–1.13× at one and five records, 0.50× at 100) and places the crossing between 5 and 20 records, but it tested only the ends; no study here localizes a single crossover point across epochs.

Finding. Toolplane *costs more to start and less per record. The registered result is a loss on small tasks and a gain on large fine-grained ones; no study here fixes a single crossover point.*

5.2 Why round trips do not explain the win

The pilot saw the mechanism first, in request IDs; Table 4 later confirmed it in token classes. This subsection keeps only the pilot's own evidence. Native-client Claude Code grouped many independent tool calls into a small number of model responses. At $N = 100$, the native arm produced 103 tool invocations but not 103 model requests. Conversely, the Toolplane arm required search, code execution, and answer-generation requests. Exact request IDs in the clean post-intervention traces give median request counts of 4 versus 6 at $N = 30$ and 5 versus 6 at $N = 100$ for native and Toolplane respectively. The Toolplane arm therefore makes more requests and still wins at $N = 100$.

The scaling term is instead model-generated tool syntax and model-visible results. Every native `get_order` requires an output-token tool block; every returned record becomes conversation material unless the client externalizes it. At $N = 100$, the native arm generated roughly 7.5K output tokens for a three-line aggregate. The Toolplane arm generated one loop and returned only the aggregate, leaving intermediate records in the sandbox. Parallel scheduling reduces native wall time but does not remove generated syntax or conversation growth.

The later payload experiment records exact request counts under the hardened harness. At 20 KB/fetch-one, the native arm batches 30 record fetches into a median of four model requests and 32 tool uses; the Toolplane arm uses a median of five model requests and four tool uses, yet costs 0.10× native. This cleanly separates request batching from payload isolation. The registered matrix shows the same split in its token classes (Table 4): generated output at small payloads, uncached input at large ones.

Finding. *Tool calls and model requests are different counts. Because the client batches calls,* Toolplane*'s advantage is less generated output and less intermediate data in context, not fewer round trips.*

5.3 A templated chain handled stepwise

The chain task follows a four-hop thread. Each fetched note names a true next order and a decoy, prompting the agent to interpret the current result before choosing the next call. Both arms are correct in all three runs, but native costs \$0.17 versus \$0.24 for Toolplane: the latter is 38% costlier on unrounded medians, with wall medians of 22.9 versus 27.9 seconds and disjoint observed ranges.

Every code run used five separate `execute_code` calls—one per node, with no execution errors. The agent did not turn the whole chain into one program; it reproduced native stepwise tool use inside a heavier interface. The fixture is template-generated and can be traversed by a hand-written regex, so this experiment measures the strategy agents selected for an adaptive-looking task, not

the theoretical impossibility of code batching. The 38% gap (1.36× on cost of pass) belongs to this July point: on cost of pass the same templated chain measured 1.07× in the October re-run on Sonnet 5, 1.05× in four runs on Sonnet 5.5, and 1.12× in the registered Haiku 5.5 matrix. A prose chain built to resist heuristic shortcuts (Section 5.7) gave the same behavior.

Finding. *On the templated chain the agent made one decision per fetched note, so the* TOOLPLANE *arm did the same stepwise work as the native arm and paid for one extra execution per step.*

5.4 Tool latency and concurrency

We added 100 ms of server-side latency to each of the 31 calls in the $N = 30$ loop. On the original await-triggered runtime, external calls dispatched only when awaited, so the recorded snippets' 31 sequential awaits added a deterministic 3.1 seconds to that execution path. End-to-end Toolplane wall medians moved from 20.9 to 23.6 seconds with overlapping ranges (unresolved at three repetitions), while native wall showed no corresponding increase, consistent with Claude Code batching independent calls. Toolplane retained the observed wall-time lead at 100 ms in that epoch.

The later eager-dispatch runtime changed the mechanism: external calls dispatch when invoked, so firing calls before awaiting them overlaps the latency — at this fixture's shape, 30 calls at 100 ms complete in 0.24 seconds through the mediated path versus 3.3 seconds serial (measured host-side, without a model). Re-running the cell on the new runtime with concurrency guidance present in the search footer, namespace manifest, and skill documentation produced wall parity with the native arm (medians 30.3 versus 28.4 seconds, ranges overlapping): the runtime could overlap, but all five recorded snippets still awaited serially inside the loop body. Adding the fan-out idiom to the `execute_code` tool description — the one surface the model always reads before writing — flipped adoption from 0/5 to 4/4 snippets on the same cell and client version, and flipped the wall result with it: Toolplane medians of 22.8 versus 28.4 seconds for the native arm, with disjoint ranges, at 29% lower cost (three repetitions per arm, counterbalanced; the native reference rows come from the same-evening control matrix).

Finding. *A runtime that can run calls concurrently helped only once the agent was shown the idiom in the tool description it reads first. The registered re-test left the wall-time difference unresolved.*

5.5 Deferred loading shrinks the unused-schema tax

A common argument for code mode is that a single facade prevents dozens of tool schemas from occupying every prompt. We varied configured MCP servers from 1 to 5 to 15, reaching 84 capabilities at the largest setting. At $N = 30$, the native-minus-Toolplane cost is non-monotone: native/Toolplane medians are \$0.20/\$0.22, \$0.20/\$0.18, and \$0.23/\$0.23 at $M = 1, 5, 15$. With three runs per cell, this is not a registry-size trend.

The trace mechanism is stronger than the noisy medians. The committed $M = 15$ init-event probe lists no loaded MCP tools; Claude Code exposes deferred stubs instead. All 18 native-arm rows across M make one client-level `ToolSearch`, load only the order tools, and select no distractor; all 36 runs are correct. At $M = 15$, none of the 82 distractor schemas is present in the probed native context. Two $M = 15$ native runs include unrelated Bash exploration, which explains much of their residual cost increase. The distractors are semantically distant, so this experiment does not test near-miss retrieval.

The registered Haiku 5.5 matrix measured the native arm at $M = 15$ against $M = 1$ with eight runs each: +3%, with an interval inside the ±10% margin but excluding zero (H8, Section 4.1). The tax is small, not absent.

Finding. *With deferred loading, fifteen configured servers cost the native arm about 3% more than one. The number of schemas configured is not the number the model sees.*

5.6 Hybrid routing is task- and scale-dependent

We tested a hybrid facade that re-exported capabilities as native MCP tools alongside search and execution. At $M = 1$ and two repetitions, the model routed single lookup and chain tasks to native tools and loops to code in every run, but economics remained task-dependent. Native/facade/hybrid medians were \$0.128/\$0.148/\$0.141 for single, \$0.205/\$0.181/\$0.155 at $N = 30$, \$0.329/\$0.180/\$0.154 at $N = 100$, and \$0.169/\$0.182/\$0.233 for the chain. Hybrid wins the loops but is worst on the chain. Neither code path invokes Toolplane's capability-search meta-tool (both may still use client-level `ToolSearch`); hybrid succeeds with one execution versus two or three under the pure facade.

At $M = 15$, all-capability hybrid exposes 87 tools rather than the facade's three. On the two measured lookup/chain tasks, it is the most expensive arm: \$0.26 for single and \$0.28 for chain, versus native's \$0.13 and \$0.19. These are two-repetition results, and loops were not run at $M = 15$; they demonstrate a failure mode, not a characterized scale curve. Loading the facade namespace also loaded 84 re-exported schemas, rebuilding the context surface the facade was intended to hide.

Selective re-export, at two repetitions per task, reduces the visible surface from 87 to five tools but does not consistently improve cost: it improves the single task relative to all-capability hybrid and is slightly worse on the chain. Follow-up name/description A/Bs are exploratory and are summarized in Appendix C; they show no detectable repair of client-mediated retrieval in the two tested domains.

Finding. *Whether an agent routes a task to a native tool or to code is its own behavior, not something the server sets. Re-exporting a few tools helped at small scale; re-exporting a whole registry did not, and renaming tools did not repair retrieval.*

5.7 October replications

Three four-run studies preceded the registered matrix. Each is reported in full in `docs/code-mode-benchmark.md`.

The shipped runtime, and a new failure. Re-running seven pilot cells on Toolplane 0.5.0 (Monty 1.1) with Claude Sonnet 5 and Claude Code 2.1.293, 24 of 24 Toolplane runs that reached the backend were correct. The discovery turn was gone: no run called Toolplane's capability search, because the facade's tool descriptions now carry call shapes, and Toolplane's wall time roughly halved at every task size. The re-run also found the study's first behavioral failure: 4 of 28 Toolplane runs abstained, answering in one turn that they had no access to order data without calling any tool. None of 21 July runs and none of 28 same-day native runs abstained. A follow-up registered on the issue tracker found no abstention on the newer models (0 of 60 Claude Haiku 5.5 runs and 0 of 40 Claude Sonnet 5.5 runs), and the confirmation matrix later recorded none in 232 Haiku runs. The follow-up ran only the 30-record cell, where Sonnet 5 itself had not abstained, so the trigger was never reproduced; it is recorded as a Claude Sonnet 5 behavior with that caveat.

A second client. On Codex CLI 0.160.0, whether the boundary exists depends on the model, because Codex gates its own built-in code mode per model. With gpt-5.5, which makes ordinary per-tool calls, the Claude Code pattern reappears: a single lookup uses 2.4 times the input, while a 30-record loop uses 25% less input and 76% less output, and a 100-record loop 92% less output. With gpt-6.1-sol, which reaches MCP tools only through Codex's own JavaScript code mode, input differs by 5% or less, walls overlap, and output is 8–24% higher for Toolplane on the small tasks: code mode calling code mode leaves little to gain and costs little. Codex on a subscription plan reports tokens, not dollars, and these are four-run cells.

A code-resistant chain and a CLI + MCP join. On Claude Sonnet 5.5, a prose chain whose notes identify the next order only by meaning (five heuristic shortcuts, pinned in tests, all walk to the wrong order) gave the same result as the templated chain: every run in both arms walked hop by hop, at 1.08× native cost. One shortcut stays open — fetch all notes in one program and read them — and no agent took it. A join task whose refund list lives in git history and whose amounts live behind MCP measured 1.22× native with a shell available — most likely because Toolplane runs explored the repository through Bash first, though they also made one capability search each — and 0.94× with the shell removed and git reached through Toolplane's CLI binding. The registered Haiku 5.5 re-test of the same join is in Section 4.1 (H7a and H7b).

6 A Conditional Empirical Envelope

The measurements sort tasks by what the model contributes at each step. Where it only carries values between tools, a program is cheaper once there are enough of them. Where the agent decides step by step, or a single call already returns everything, there is little glue to remove and the fixed cost of the program is not repaid. The measurements support a decision envelope for the tested system, not a predictive law. Table 10 states the observed direction and the mechanism that must hold for it to transfer; Figure 8 summarizes the same logic as a checklist.

The envelope can be read as a sequence of questions:

1. Does the server expose the measured bulk retrieval operation? The native-client path avoids Toolplane's programmable layer; at large payloads it may still perform context-decoupled work through its own file/code path.

2. Does the agent handle the task one observation at a time? In the measured templated and prose chains, native registration avoids repeated execution envelopes. The prose chain was built so that no heuristic we tried can shortcut it, though one program that fetches every note and hands them all to the model remains possible; no agent tried it.

3. Otherwise, are there many fine-grained calls with model-visible intermediate results? Toolplane can prevent generated call syntax and payload ingress from scaling with the task.

4. Are calls slow and independent? The answer depends on whether the code runtime can fan them out as effectively as the client — and on whether the fan-out idiom is taught where the agent reads it.

A task-specific server-side filter or aggregate should further favor the native path under the cost model, but that branch is a prediction: this study measures bulk retrieval, not server-side answer computation.

Most of this envelope's directions were pre-registered and confirmed on Claude Haiku 5.5, but on the same order-store domain on which the pilot discovered them. Predicting winners on held-out domains remains untested, so "envelope" means a synthesis of controlled observations, not validated automatic routing or a claim of optimal interface selection.

Table 10: Observed decision envelope. Registered ratios are Claude Haiku 5.5 cost of pass, TOOLPLANE over native, except the session row (Claude Sonnet 5.5). Directions should be tested before use as a routing policy.

Condition	Observed preference	Mechanism
Single lookup or about five fine-grained calls	Native (registered: 1.11–1.13×)	Fixed search/code overhead cannot amortize
20–30 small fetch-one records	Toolplane (registered at 30: 0.82×)	20 was run but not tested (0.93×); the July Sonnet 5 ranges at 20 and 30 overlapped
Many fine-grained fetch-one calls, especially with large model-visible results	Toolplane (registered: 0.50× at 100; 0.020× at 20 KB)	Generated call syntax and intermediate payload stay outside model context
Native bulk endpoint	Native (registered: 1.05–1.31×)	One domain call removes fine-grained composition; large results may enter the client's own file/code path
Chain handled stepwise, templated or prose	Native (registered: 1.12×)	Toolplane adds one execution envelope per agent-selected stage
Slow independent calls, serial sandbox awaits	Native pressure (pilot)	Client batches native calls while the runtime pays approximately $K\ell$
Slow independent calls, fan-out idiom taught	Toolplane on cost; wall time unresolved (registered)	Eager dispatch collapses the sandbox path to about ℓ; the pilot's wall-time win did not replicate with an interval
Large, semantically distant registry with deferred loading	Small native schema tax (registered: +3% at $M = 15$)	Unused schemas mostly stay out of model context
Related queries over already loaded data	Session total: Toolplane (registered: −40%); each reuse turn: native (+35%)	The saving is the load; every reuse turn needs an extra executor request
Client that already runs its own code mode	Neither (Codex gpt-6.1-sol, four runs)	Code mode calling code mode leaves little to save

7 Related Work

Executable actions and context-decoupled agents. CodeAct replaces constrained JSON/text actions with executable Python and reports improved task success across models [22]; DynaSaur lets agents create, compose, and reuse programmatic actions dynamically [17]. Felendler et al. provide the closest direct MCP architectural comparison: they formalize context-coupled and code-execution MCP, measure tokens, latency, turns, and task quality across servers and models, and develop a security taxonomy [8]. LOCA-bench independently controls environment-driven context growth while holding task semantics fixed and evaluates programmatic tool calling among several context management strategies [27]. CoDA isolates planner and executor contexts, while ATLAS combines progressive MCP loading with programmatic orchestration for learned small-model agents [13, 10]. These works establish the representational and context-management value of executable or isolated action paths. We neither claim the first direct-versus-code comparison nor contribute a security taxonomy. Our narrower contribution is billed-cost and trace-level localization across interaction, payload, granularity, discovery, latency, and persistent-session axes in one production client.

Production systems. Anthropic reports both a roughly 38% billed-input reduction on a 75-tool project-management benchmark and roughly 8% higher cost with unchanged scores on sequential τ^2-bench tasks making one or two calls per turn [3]. Cloudflare reports large static schema-token reductions for a code-mode API facade [4], following its earlier proposal to convert MCP tools into a typed API executed in a sandbox [21]; Anthropic's engineering reports motivate the same context-isolation pattern [1, 2]. The goose agent ships an open-source code mode, and its maintainers report 30% fewer tokens on one task while noting that for simple single-tool tasks the overhead is not worth it [20]. Codex CLI ships its own JavaScript code mode and enables it per model; in our replication, models with it routed every MCP call through code, while gpt-5.5 made plain tool calls (Section 5.7). Workload-dependent gains and losses are therefore already publicly documented. Our contribution is not the first reported reversal; it is an independently reproducible, multi-axis measurement with committed raw artifacts and explicit counterexamples.

Tool retrieval and dynamic catalogs. ToolLLM and AnyTool study selection from very large API collections [19, 5]. MCP-Zero and SING retrieve capabilities from evolving model intentions while minimizing schema exposure [7, 23]; MemTool manages tool context across long conversations [14]. CodeNav lets agents search and import code rather than consume registered tool descriptions [11]. These works primarily optimize recall, success, or schema tokens. Among the works we reviewed, none reports discovery as incremental production-client model

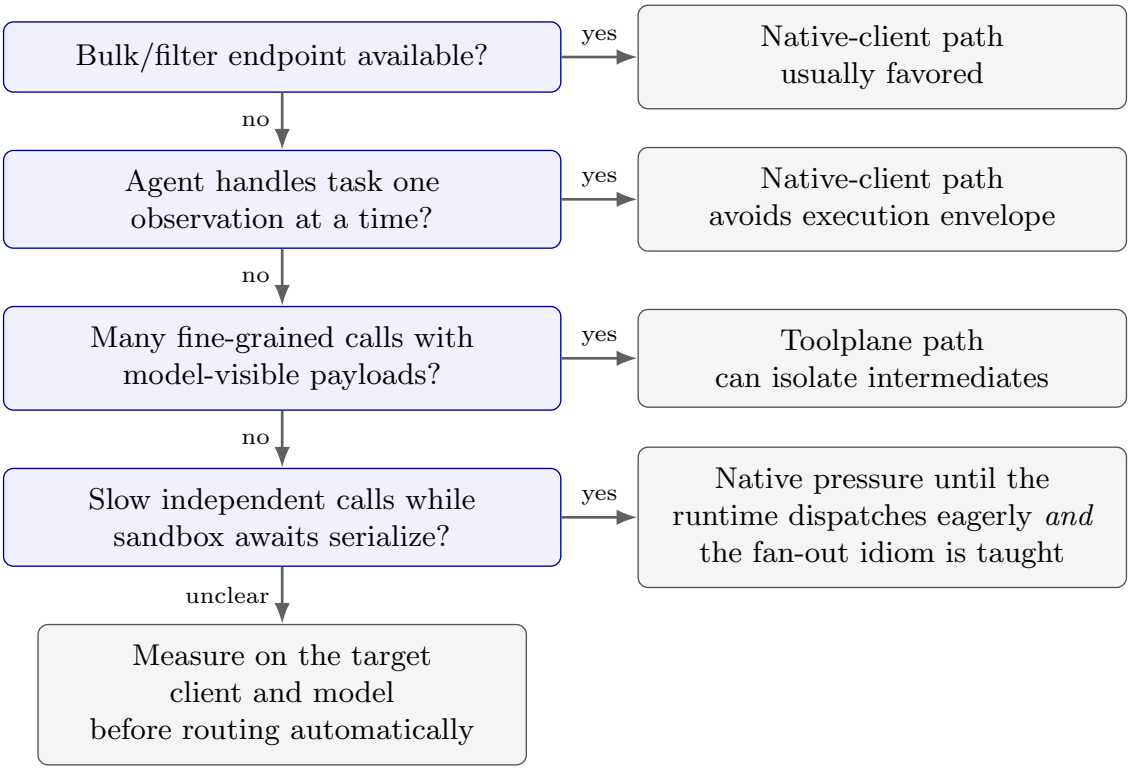


Figure 8: Checklist form of the observed envelope. The diagram is not an automatic router: each branch encodes a mechanism observed in this Claude Code/Toolplane case study (registered on Claude Haiku 5.5) and requires validation before transfer to another client, model, runtime, or workload.

requests and prices those requests in a controlled direct-versus-code comparison. Our registry and hybrid results also show why catalog size alone is an insufficient independent variable when the client defers loading.

Agent interfaces, caching, concurrency, and economic evaluation. SWE-agent demonstrates that agent-computer interface design materially changes model behavior [26]; our object is a specific economic interface boundary rather than the first interface-design result. Prompt caching can reduce agent cost and latency substantially [15], making it essential to price cache reads rather than compare raw context sizes. Asynchronous function-calling work studies latency benefits from parallel execution [9], while TPS-Bench and BFCL evaluate planning, scheduling, parallel calls, and tool-use quality [24, 18]. These evaluations treat parallel execution as a capability to be measured. Our latency A/B isolates a further variable they do not price: whether the model actually writes the concurrent idiom. A verified runtime capability produced no measured benefit until the idiom moved into the executor's tool description (Section 5.4) — an adoption effect in the spirit of SWE-agent's interface result, at the granularity of a single description sentence. Cost-of-Pass and AI Agents That Matter argue for economically grounded evaluation rather than accuracy-only rankings [6, 12]. We apply that principle to the interface crossover; we do not claim the first cost-aware agent benchmark.

8 Threats to Validity

Few clients and models, one code-mode implementation. Claude Code's ToolSearch, native batching, built-ins, prompt caching, and result-file externalization are central causal actors, and each model's routing and program-writing behavior is inseparable from the client treatment. The registered evidence covers Claude Haiku 5.5 (cost envelope) and Claude Sonnet 5.5 (sessions) in one Claude Code version. Codex with two models is a four-run replication reporting tokens rather than dollars. Claude Sonnet 5 abstained in October but not in July under the same model string, while the runtime, the facade's descriptions, and the client had all changed, so a behavior can appear without any change on the measured side of the interface. An independent code-mode implementation is untested.

Synthetic and narrow domains. Most economic tasks use formula-generated order records. This gives exact correctness and controllable payloads, but aggregation naturally favors code and does not represent a workload distribution. The 2 KB and 20 KB payload levels are unused synthetic detail strings, not samples from a real domain; they isolate payload ingress rather than payload semantics. The shipment domain is used only to challenge

a discovery naming hypothesis. The strongest further validation would add genuinely different document/search and operational API domains, including quality-sensitive tasks where both arms can fail.

Small samples and multiplicity. The registered studies use eight runs per arm (four in the 20 KB cells) with bootstrap intervals; pilot cells have two to four, and hybrid cells often two. The pilot evolved through many interventions and carries multiple-comparison and researcher-degrees-of-freedom risk; the registrations exist to bound it. The confirmation matrix tests 22 rows without a multiplicity correction, and some of its cells are nearly deterministic, so a single outlier can decide an interval. Two procedural lessons are recorded for later registrations: run one discarded warm-up per block, because the first run after a cache expiry pays cache creation, and test equivalence margins first.

Native baseline and API design. The bulk intervention repairs an important weak-baseline concern, but server-side filters and aggregates remain unmeasured. A server-side operation that returns exactly the requested answer should usually favor native tools; testing it would measure the boundary rather than overturn the model. Large bulk results also trigger a client-specific file/Bash path. Consequently, those cells compare Claude Code's native MCP + result-file + Bash/Python path with Toolplane-mediated execution; they do not isolate a pure "direct, no-code" paradigm.

Cost and correctness. Most cells are fully correct, which makes cost comparison clean but says little about quality. Three failures are the exception, and each is a model behavior the cost model does not contain: Claude Sonnet 5 abstaining behind the generic facade (4/28), Claude Haiku 5.5 degenerating after 30 native fetches of 2 KB records (0/8), and Claude Haiku 5.5 never searching for the facade when a shell was available (2/8). Cost of pass prices them in, so two registered verdicts hold through them; we report those as correctness results. Tasks where both interfaces can fail on quality remain untested.

Implementation versus paradigm. Fixed discovery cost, runtime syntax support, and agent-authored retry behavior are properties of the measured Toolplane version. The latency cells further show results moving with runtime dispatch semantics (await-triggered versus eager) and with where concurrency teaching is placed; other code-mode systems may search, batch, sandbox, expose types, or document concurrency differently. Conversely, native performance depends on whether the client defers tools and externalizes large results. We distinguish these mechanisms rather than attributing all observations to an abstract paradigm.

Temporal prices and latency. Dollar values depend on provider pricing, caching policy, client version, and run date. Raw token classes exist for all rows; exact request IDs and complete transcripts exist for later harness epochs. These permit partial repricing and trace reanalysis. Wall-clock variance includes remote model service behavior; cross-version wall times are not compared.

Security is out of scope. Code execution expands the attack surface and changes auditability. The experimental runtime is sandboxed and capabilities traverse host policy and audit gates, but this paper does not evaluate adversarial safety. Felendler et al. provide a dedicated taxonomy and defense analysis [8]. Performance evidence must not be read as a security endorsement.

9 Reproducibility and Research Practice

The repository publishes the benchmark driver, deterministic fixtures, validators, and raw JSON for all reported runs. For matrices on the hardened harness it also publishes complete redacted transcripts, classifiers, exact model request accounting, and frozen-code provenance. Later rows stamp the built wheel, fixture set, base harness, and experiment-specific harness hashes. The paper's evidence map names the primary artifact for each table and marks weaker epochs.

Both October studies were pre-registered: the plan, hypotheses, decision rule and analysis script were committed before any data, and the results documents report every verdict, including the reversed and unresolved ones, with independent recomputation from the raw rows.

We also retain failures of the research process. A contaminated paid matrix is named and excluded rather than deleted. Negative product hypotheses—the templated chain, all-capability hybrid, selective re-export, naming intervention, and compounding-session story—are reported alongside wins. Post-hoc changes are explicit: one correctness validator was semantically rescored without changing answers, and longitudinal per-turn token fields were replaced from verbatim result events after an accounting error. The shipment naming runs' dirty, not-fully-hash-covered harness provenance is retained rather than presented as clean. None of these corrections changes the primary tables in this paper.

This transparency does not substitute for independent replication, but it makes disagreement testable. The artifact is MIT-licensed. It does not yet include a machine-readable pricing schedule, so re-pricing the token classes at other rates is left to the reader.

10 Conclusion

What separates the two interfaces is what the model does at each step: carry values, or decide. Native-client MCP

orchestration and Toolplane-mediated execution are not ordered by a single notion of task complexity. In the measured system, Toolplane pays a fixed discovery and program-generation cost, then avoids model-generated per-call syntax and model-visible intermediate payload. A pre-registered matrix on Claude Haiku 5.5 confirms the shape the July pilot found: native registration wins a single lookup and a five-record loop by 11–13%; Toolplane halves cost at 100 fine-grained calls and is 50 times cheaper when those calls return 20 KB records. A native bulk API reverses the winner, and agents walk chains step by step, so the native path wins those too, even on a chain written to resist shortcuts. Request counts barely differ between arms: the advantage is output tokens at small payloads and uncached input at large ones. Client mechanisms erase textbook arguments on both sides: deferred loading leaves only a small schema tax, while result externalization gives the native arm its own context-decoupled file/code path. Slow calls make the boundary second-order: the pilot's wall-time win needed both eager dispatch and a taught fan-out idiom, and the registered re-test left it unresolved.

The persistence study sharpens the data-ingress mechanism. On Claude Sonnet 5.5 a Toolplane session lowers five-turn cost by 40% and peak context by 38%, but the gap opens on the initial load; each reuse turn then costs about a third more than native conversation reuse, because the session needs an executor call before it can answer. Against a sessions-off Toolplane arm the live session is 7% cheaper, but that arm refetched inside the sandbox instead of using the result store, so the store's economic contribution is not isolated.

The practical conclusion is conditional, and it follows from what the model contributes at each step. Where the model only carries values between tools, it is glue, and in the measured system that glue cost less in a program: prefer Toolplane when many fine-grained calls expose intermediate results, especially large ones, and when the client does not already run its own code mode. Where there is little to carry, or the agent decides step by step, keep the model in the loop: prefer the native-client path for a handful of calls, when a sufficient bulk operation exists, or when agents handle a chain stepwise. "Use the model for judgment, not for glue" is a mnemonic for these measurements, on one domain and a small set of clients and models. It is not a decision rule, and not a law. Treat discovery, client behavior, and runtime concurrency as first-class parts of the system: the study's three failures are model behaviors at the interface, not runtime errors. More broadly, Toolplane is useful when a small programmable surface over MCP, CLI, and host/Python capabilities is worth an extra execution layer; this paper identifies conditions under which that trade-off is economically visible. Held-out domains and further clients are the next step from this case study to a general result.

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Appendix

A Evidence Tier Map

Table 11: Evidence tiers used for interpretation.

Tier	Evidence	Role in the paper
Primary	Registered confirmation matrix (Claude Haiku 5.5); registered persistence study (Claude Sonnet 5.5)	Anchor the cost, context, and session conclusions
Secondary	July payload × granularity and longitudinal pilot; July crossover, templated chain, latency and latency-adoption A/B, registry size, hybrid routing; October four-run replications (runtime re-run, Codex, prose chain, CLI join)	Mechanism probes and replications; useful for the envelope but not population estimates
Exploratory	Naming/description A/Bs, pre-hardening pilots, the confirmation matrix's two-run Sonnet 5.5 slice, and corrected historical rows	Hypothesis generation and provenance, not standalone support for headline claims

B Primary Artifact Map

Table 12: Committed source artifacts for the manuscript's principal results.

Result	Artifact
Original single/$N = 30$ and $N = 100$	`run-20260708-123900.json`; `run-20260708-124546.json`
Registry $M = 1/5/15$	`run-20260709-114215.json`
Discovery intervention	`run-20260709-221840.json` and transcripts
Crossover, templated chain, latency	`run-20260709-235016.json` and transcripts
Latency control and adoption A/B	`run-20260719-023232.json`; `run-20260719-025053.json` and nine transcripts
Hybrid and curated hybrid	`run-20260711-001559.json`; `run-20260711-002646.json`; `run-20260711-014100.json`
Naming/description A/B	`run-20260712-014825.json`; `run-20260712-015009.json`; `run-20260712-015759.json`; `run-20260712-100935.json`; `run-20260712-101102.json`
Payload × granularity	`run-20260713-040737.json` and 36 transcripts
Longitudinal pilot	`longitudinal-20260713-095957.json` and eight transcripts
Confirmation matrix	`run-20261010-115818.json` through `run-20261010-142344.json` (eight files) and transcripts; `PREREG-confirmation-2026-10.md`; `RESULTS-confirmation-2026-10.md`
October runtime re-run; abstention follow-up	`run-20261008-012307.json`; `run-20261008-082359.json`; `run-20261008-091632.json`
Codex replication	`run-20261008-202709.json`; `run-20261008-203351.json`
Prose chain; CLI + MCP join	`run-20261009-131413.json`; `run-20261009-132058.json`
Persistence study, Sonnet 5.5 slice, Monty 1.1 snapshots	`longitudinal-20261010-191310.json` and 24 transcripts; `run-20261010-192444.json` and 32 transcripts; `PREREG-persistence-2026-10.md`; `RESULTS-persistence-2026-10.md`

C Naming and Description A/B Details

The follow-up retrieval A/Bs were exploratory because model-generated search queries vary by arm and the shipment rows retain dirty, not-fully-hash-covered harness provenance. On orders ($n = 11$ per arm), first-search hits are 11/11 native, 4/11 control, 6/11 name, and 4/11 description; name versus control has Fisher $p = .67$. On shipments ($n = 9$), first-search hits are 9/9 native, 2/9 control, 3/9 name, and 3/9 description, with $p = 1.0$. The narrow conclusion is

that server-side name or description changes did not detectably repair client-mediated retrieval in these two domains. The client-set server qualifier and competition with built-in tools remain plausible mechanisms, not isolated causes.

D Longitudinal Correction and Reset Revalidation

Disclosed exceptions. Several exceptions are retained and disclosed. One discovery-tax matrix was contaminated by a mid-run facade change; it is committed under a `mixed-facade` name and excluded. An early correctness validator was semantically rescored from committed answers without changing answers or usage. The shipment naming runs recorded `git_dirty=true`; their wheel is anchored and fixture hashes match the subsequently committed bytes, but the then-uncommitted `bench/run.py` was not hash-covered. Finally, the longitudinal harness initially differenced token usage fields that were already per-turn. Those four fields were deterministically replaced from the committed result events; normalized comparison showed no change to cost, context, answers, fixture calls, or published table values. No paid run was repeated for these corrections.

The longitudinal correction. The recorded longitudinal result events expose per-turn usage, but the first analysis treated those counters as cumulative and subtracted adjacent turns. This produced negative values in four token fields. The committed artifact replaces only those fields from each transcript's verbatim result event. A normalized before/after comparison established that billed cost, peak context, answers, correctness, request counts, and fixture calls did not change. The economic analysis in Section 4.2 uses billed cost and request-context values unaffected by the correction.

The reset predicate was also strengthened after the paid run. The final predicate requires a sole, zero-argument reset in the first execute-bearing assistant event, a matching non-error tool result, and calculation in a later assistant event. It independently revalidates all four committed reset-turn transcripts of the pilot. The reset turn is contract evidence and is excluded from the five-turn economic comparison.

In the pre-registered study the same predicate credited 6 of 8 Toolplane resets. The other two transcripts show `return await reset_session()` as the sole statement of its own execution, followed by a full refetch and a correct answer; the predicate accepted only the bare-statement form. Reset is reported, not tested, so no verdict depends on it, and the registered figures stand as recorded. The predicate was fixed afterwards for future runs; replayed over these transcripts it credits 8 of 8.